

ATOMOD

User guide

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Version 0.6.0

July 7, 2026

Installation

Before you begin, check the ATOMOD systems requirements, to avoid any issues during the process. Ensure you have the following:

- PYTHON:
 - Requires Python 3.12 or an earlier version (e.g., 3.11, 3.10) to avoid compilation errors.
- DISK SPACE:
 - At least 7 GB of free space on your partition.
- SYSTEM SOFTWARE (Required):
 - `git` (to clone the ATOMOD repository)
 - `python3-venv` (to isolate Python packages)
 - `python3-pip` (to install dependencies)
 - `wget` (to download installer `install.py`)

On Ubuntu / Xubuntu / Debian, you can install everything using:

```
sudo apt update && sudo apt install git python3-pip python3-venv wget
```

Windows

Installation process on windows:

- 1 - Open a PowerShell (Fig.1(1-2-3))
- 2 - Navigate to the directory of your choice (e.g., `cd ~/Desktop` or `cd ~/src`)(Fig.1(4)).
`cd C:\Users\herve\Desktop\`
- 3 - Download the installation script `install.py`.
`wget https://github.com/hbulou/ATOMOD/releases/download/v0.1/install.py`
- 4 - Run the installation (Fig.1(5)). NOTE: You must have at least 7 GB of available disk space.
`python.exe install.py`
- 5 - Once the installation is complete, go to the ATOMOD directory (Fig.1(6)).
`cd ATOMOD`

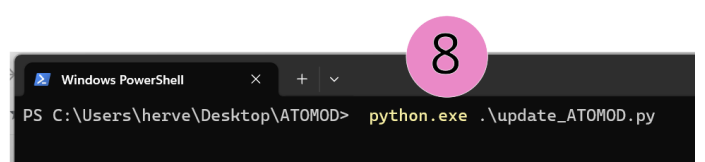
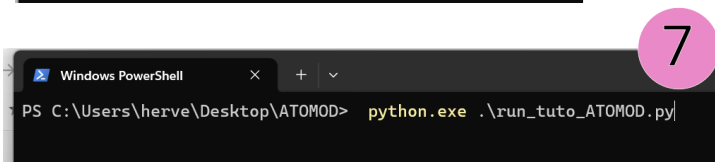
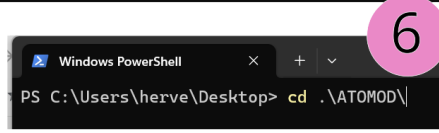
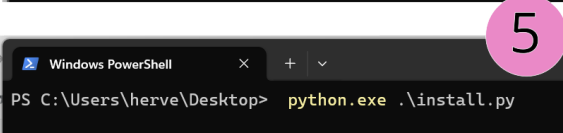
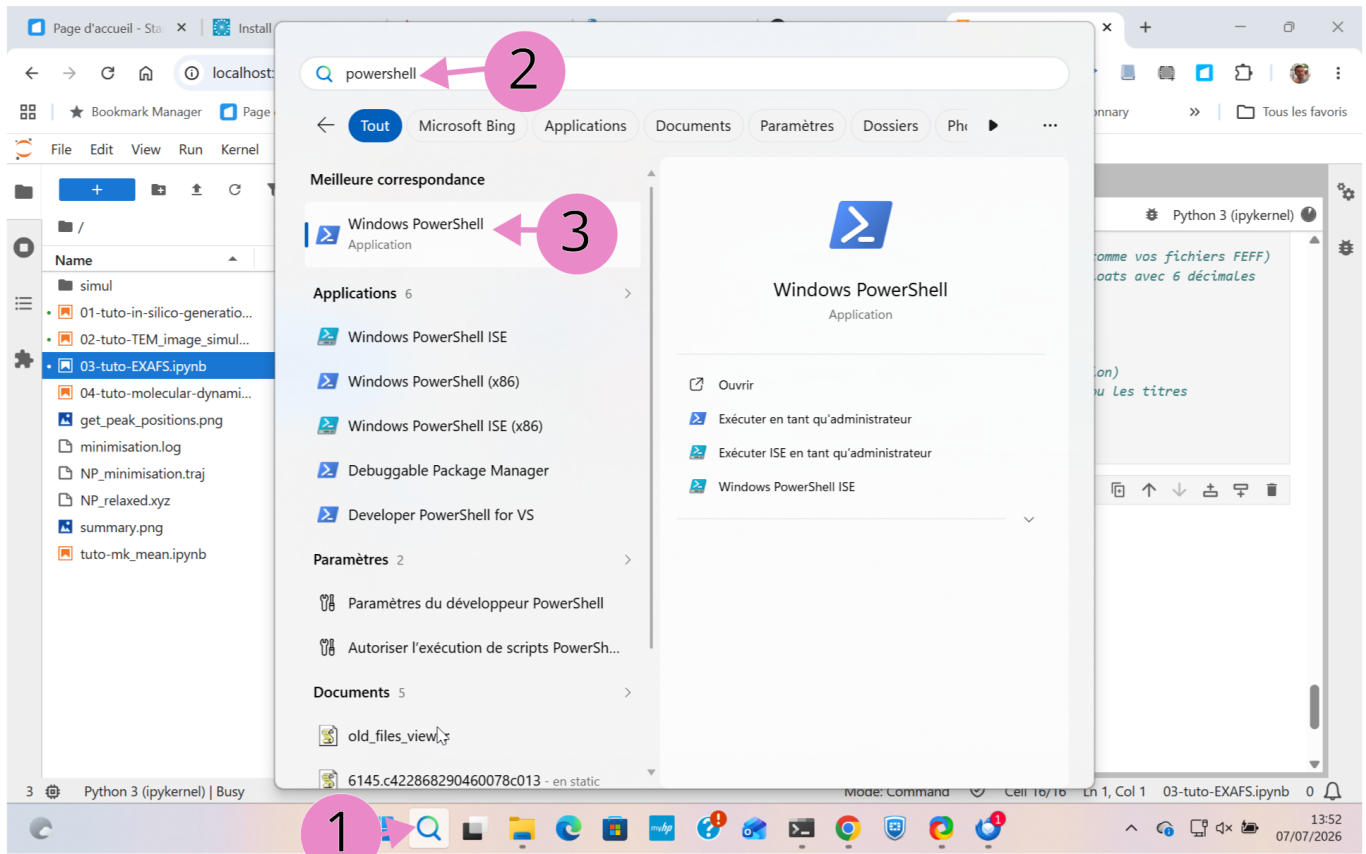


Figure 1: Screenshots of the various stages of installing ATOMOD on Windows.

Linux

Installation process on linux:

- 1 - Open an X terminal
- 2 - Navigate to the directory of your choice (e.g., `cd ~/Desktop` or `cd ~/src`)(Fig.2(a)).
`cd Desktop`
- 3 - Download the installation script `install.py` (Fig.2(b)).
`wget https://github.com/hbulou/ATOMOD/releases/download/v0.1/install.py`
- 4 - Run the installation (Fig.2(c)). NOTE: You must have at least 7 GB of available disk space.
`python3 install.py`

- cd ATOMOD

```
(d) Requirement already satisfied: pure-eval in ./ATOMOD/venv/ATOMOD/lib/python3.12/site-packages (from stack_data=>0.6.0->ipython=>6.1.0->ipywidgets) (0.2.3)
Downloading ipywidgets-8.1.8-py3-none-any.whl (139 kB)
Downloading jupyterlab_widgets-3.0.16-py3-none-any.whl (914 kB)
914.9/914.9 kB 12.1 MB/s 0:00:00
Downloading widgetsnbextension-4.0.15-py3-none-any.whl (2.2 MB)
2.2/2.2 MB 11.9 MB/s 0:00:00
Installing collected packages: widgetsnbextension, jupyterlab_widgets, ipywidgets
Successfully installed ipywidgets-8.1.8 jupyterlab_widgets-3.0.16 widgetsnbextension-4.0.15
Collecting py3Dmol
  Downloading py3dmol-2.5.5-py2.py3-none-any.whl.metadata (2.1 kB)
Downloading py3dmol-2.5.5-py2.py3-none-any.whl (7.2 kB)
Installing collected packages: py3Dmol
Successfully installed py3Dmol-2.5.5
✔ Installation complete!
bulou@pc-hervecal:~/Desktop$ cd ATOMOD/
bulou@pc-hervecal:~/Desktop/ATOMOD$
```

Figure 2: Screenshots of the various stages of installing ATOMOD on Linux.

ATOMOD Update

To update ATOMOS, simply run the `update ATOMOD.py` script (Fig.1(8)):

```
python3 update ATOMOD.py
```

Tutorials

Four tutorials are available in the `./doc/tutorials` directory:

- 01-tuto-in-silico-generation.ipynb, describing the procedure for creating a nanoparticle *in silico*,
- 02-tuto-TEM_image_simulation.ipynb, describing the procedure for simulating a TEM image,
- 03-tuto-EXAFS.ipynb, describing the procedure for simulating the EXAFS spectra of a nanoparticle,

- `04-tuto-molecular-dynamics.ipynb`, describing the procedure for performing classical molecular dynamics (MACE force fields).

To run the tutorials, the simplest method is to use the `run_tuto_ATOMOD.py` file located in the `ATOMOD` directory ((Fig.1(7)):

```
python3 run_tuto_ATOMOD.py
```

Note regarding `03-tuto-EXAFS.ipynb`: the tutorial was developed for use with `FEFF` installed outside the `ATOMOD` distribution; therefore, this tutorial will not work if you do not have `FEFF` installed on your computer.